Mid Exam

Sorna MAM

Peripheral device →

Peripheral device are any device that can be attached to a CPU.
Ex- Hard disk, mouse, printer.

Interface →

The interface is the combination of hardware and software needed to link the CPU to the peripherals and to enable them to communicate with the CPU.

Main funtion of interface →

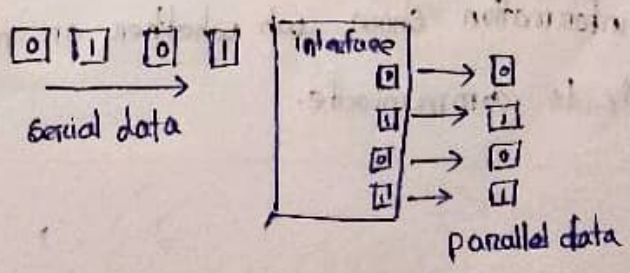
i) Buffering.
ii) Converting data to serial and parallel form.
iii) Converting data to digital and analogue form.
iv) Voltage conversion.
v) Protocol conversion.
vi) Handling of status signal.

Buffer →

Buffer is an area of RAM within the interface which temporarily store the data while in transit between the processor and the peripheral.

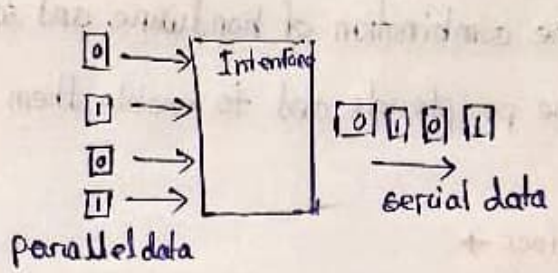
Serial data →

Serial data transmission is when data is transmitted along a communication channel one bit after another in sequence.
It is slow but efficient over long distance. Serial interface use serial data transmission



Parallel data →

Parallel data transmission is when transmit several bits of data simultaneously across a series of parallel channels (16/32 bit at a time).
It is fast and suitable for short distance. Parallel Interface use parallel data transmission.



DAC (Digital to Analogue Converter) →

An interface that has the ability to change the digital signal from computer to analogue signal that other device can understand is called DAC.

ADC (Analogue to digital converter) →

An interface that has the ability to change the analogue signal from other device to digital signal that computer can understand is called ADC.

Protocol conversion →

A protocol conversion means ensuring that the protocols used by the peripheral can be understood by computer it is attached to and vice versa

Status signal →

Status signal information show ~~wh~~ whether or not a peripheral device is ready to communicate.

Buffer and Spoolers →

[Def in previous page]

Why use buffer →

- Compensate for the difference in operating speeds between CPU and its peripheral
- Reduce the frequency with which the CPU is interrupted to deal with input

Spooler →

A spooler is a software program that manages the process of sending data to a device ~~in an organized~~, typically a printer or other output device in an organized and efficient manner.

Buffer vs spooler →

Buffer	Spooler
1. Handle speed difference between the CPU and peripheral device	1. Manage multipul task for peripheral, ensuring orderly execution.
2. Temporarily holds data for smooth transfer.	2. Queues multiple task and process them sequentially.
3. Used in keyboard, hard drives, network card to prevent data loss due to speed.	3. Used in printer to manage multiple task efficiently.
4. Smaller in size	4. Larger in size.
5. transfer data quickly	5. If CPU is busy spooling can be slowlen.

Interfacing speed measured in Megabit per second.

Types of computer →

Computer

Mainfram computer
- largest, most powerful
- high speed
- large data word
- Massive memory
- Used in - Military defense control, IBM4381

Super computer
- fastest and more powerful than Mainfram
- Used by largest firms, Cray Y-MP/832

Mini computer
- Run slowly
- works with smaller data
- Small memory
- Used in scientific research

Micro Computer
- Small computer
- CPU - single microprocessor
- Used personal, desktop, laptop

Microprocessor datatype →

1. Bit - ~~0/1~~ Represant by 0/1
2. Byte - 8 bit of data
3. Word - Data that is handle by microprocessor at a time
4. Signed and unsigned binary integer
5. BCD - Represented by four bit binary number.
6. ASCII - Represented by an integer
7. Floating point number.

Representation of ~~sig~~ unsigned binary number →

1. 92

$\Rightarrow (92)_{10}$

$\Rightarrow (1011100)_2$

7 bit binary magnitude of decimal number 92.

2. find rang of 5 bit unsigned binary number

$\Rightarrow 0$ to (2^n-1)

$\Rightarrow 0$ to (2^5-1)

$\Rightarrow 0$ to 31

Representation of signed binary number →

1. Sign-Magnitude

$\Rightarrow -(2^{n-1}-1)$ to $(2^{n-1}-1)$ [Range]

2. 1's complement

$\Rightarrow -(2^{n-1}-1)$ to $(2^{n-1}-1)$

Zero has two representation $-0, +0$

2. 2's complement

$\Rightarrow -(2^{n-1})$ to $(2^{n-1}-1)$

Zero has one representation $+0$

What is the range of signed value that store 20 bits?

$\Rightarrow -(2^{20-1}-1)$ to $(2^{20-1}-1)$

$\Rightarrow -524287$ to $+524287$

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Floating point number →

- Have three parts component → sign, exponent, mantissa

Example -

negative ← $\boxed{-}\boxed{25} \times 10^{\boxed{-2}}$ → Exponent

mantissa

- Represented as a normalized binary fraction raised to a power of 2.

Conversion → ***

Example- Convert 2.625 to our 8 bit floating point format.

S	Exponent	Mantissa
1 bit	3 bit	4 bit

Integral part → $2 \Rightarrow (10)_2$

$0.625 \Rightarrow (101)_2$

$0.625 \times 2 = 1.25$	1
$0.25 \times 2 = 0.5$	0
$0.5 \times 2 = 1.0$	1

$(2.625)_{10} = (10.101)_2$

Normalize →

$(10.101)_2 \times 2^0$

$\Rightarrow 1.0101 \times 2^1$

Mantissa → 0101

Exponent → $1+3 = 4 = 100$

sign bit → 0

Note- for 8 bit bias-3
for 32 bit bias-127
for negative exponent sub [sign bit 1]
for positive exponent add [sign bit 0]

0	1	0	0	0	1	0	1
Sign bit	Expo			Mantissa			

Example- Convert −4.75 to 8 bit

$(4.75)_{10} = (100.11)_2$

Normalize ⇒ $100.11 = 1.0011 \times 2^2$

Mantissa ⇒ $(0011)_2$

Exponent ⇒ $2+3 = 5 \Rightarrow (101)_2$

1	1	0	1	0	0	1	1

s Exp Mantissa

Example- Convert 0.40625 to 8 bit.

$(0.40625)_{10} = (0.01101)_2$

Normalize $\Rightarrow (0.01101)_2 = 1.101 \times 2^{-2}$

Mantissa $\Rightarrow 1010$

Exponent $\Rightarrow -2+3 = 1 \Rightarrow (001)_2$

0	0	0	1	1	0	1	0	X

Example- Convert -12.0 to 8 bit.

$(12.0)_{10} = (1100)_2$

~~Normalization~~

Normalize $\Rightarrow 1100.0 = 1.1000 \times 2^3$

Mantissa $\Rightarrow 1000$

Exponent $\Rightarrow 3+3 = 6 = 110$

1	1	1	0	1	0	0	0

Example- Convert 1.7 to 8 bit

$(1.7)_{10} = (1.1011\ldots)_2$

Already normalize - 1.1011×2^0

Mantissa - 1011

Exponent - $0+3 = 3 \Rightarrow (011)_2$

⇒

0	0	1	1	1	0	1	1

Example - Convert -1313.3125 to IEEE 32 bit floating point format

$(1313.3125)_{10} = (10100010000.0101)_2$

Normalize ⇒ $(10100010000.0101)_2 = 1.0100100000101 \times 2^{10}$

Exponent ⇒ $127 + 10 = (137)_{10} = 10001001$

Mantissa ⇒ 01001000001 01

1	1	0	0	0	1	0	0	1	0	1	0	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0

s — Expo — Mantissa

Note: Sign bit ⇒ 1 bit

Exponent ⇒ 8 bit

Mantissa ⇒ 23 bit [Empty will fill with 0]

Example - Convert 0.1015625 to IEEE 32-bit floating point

$(0.1015625)_{10} = (0.0001101)_2$

Normalize ⇒ $(0.0001101)_2 = (1.101 \times 2^{-4})$

Mantissa ⇒ 101

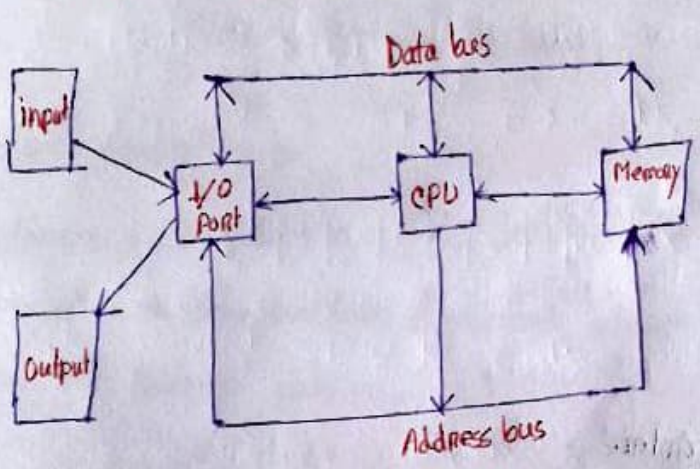
Exponent ⇒ $127 - 4 = 124 = (01111011)_2$

0	0	1	1	1	1	0	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

IEEE 64 bit →

- 1 bit for sign bit (0/1)
- Exponent (11 bit)
- Mantissa (52 bit)
- Bias (1023)

Microcomputer structure →



i) CPU
ii) Memory
iii) Input/output
iv) Buses — Address bus, Data bus, Control bus

self described

Register →

There are three type of register.

1. General purpose Register.
2. Specific purpose Register.
3. Memory register.

General purpose register →

- 6 general purpose register (8085) store 8 bit data.
- B, C, D, E, H, L. Combined BC, DE, HL to perform 16 bit operation.
- Use to store or copy temporary data

Specific purpose register →

Accumulator-

The accumulator is a 8-bit register that is the part of arithmetic and logic (ALU) unit which store result after performing arithmetic and logic operation.

Flag register-

Sign flag-

Zero flag-

Parity flag-

Carry flag-

Overflow flag-

Memory Register-

Program counter, stack pointer.

Address bus →

- Consist of 16, 20, 24, 32, 36 parallel unidirectional signal line.
- The number of the location that CPU can address is determind by the number of address lines.

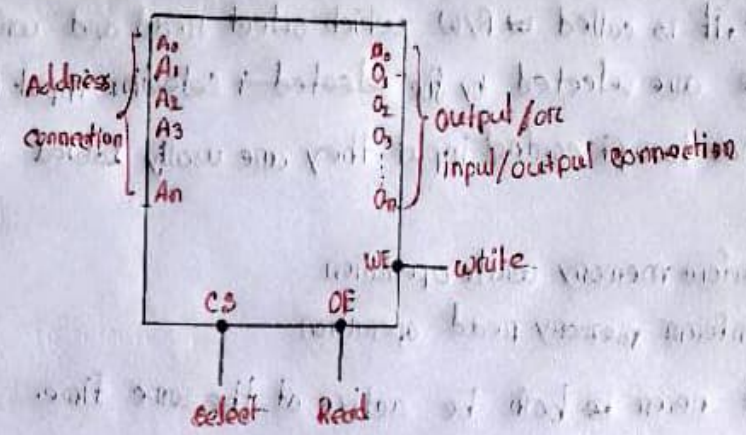
Data bus →

- Consist of 8, 16, 32 parallel bidirectional signal line

Control bus →

- Consist of 4 to 10 parallel signal lines.

Memory pin connection →



Address connection -

Address input select a memory location within the memory device. They are labeled from A_0, the least significant address input to A_n. A 1K memory device has 10 Address pins ($A_0 \cdots A_9$)

Data connection -

Data connection are the point at which data are entered for storage and extracted for reading. A 16K×1 is a memory device containing 16 K 1-bit memory location. 64K×4 memory device are listed as 256 K device.

Selection connection -

Each memory device has an input that select or elable the memory device. This are called chip select (CS), chip enable (CE) or simply select.

Control connection -

The control input most often found on ROM is the Output enable (OE) or gate (G) connection which allows data to flow out of the output data pins of ROM. If OE and CS are both active, then the output is enabled. If OE is inactive the output is disable at high impedance state.

A ~~Ram~~ RAM memory device has either one/two control input. If there is one input, it is called ~~w~~ R/W which select read and write operation when the device are selected by the ~~selected~~ selection input (cs).

If RAM has two pi control input they are usally lablod (WE)/(W) and (OE)/(G).

WE → Perform memory write operation

OE → Perform memory read operation

They must never be both be active at the same time.

Address decoding →

- Attach memory device to microprocessor
- Necessary to decode the address from microprocessor to make the memory function at unique section.
- Without an address decoder only one memory device can be connected to the microprocessor witch would make it virtually useles.

Memory array design →

Memory array design means interconnecting several memory chips.

Main Memory Array design →

Memory array design means interconnecting several memory chips.

To connect a microprocessor to ROM/RAM chips, three address-decoding techniques are commonly use-

→ Linear, decoding
→ full / Partial decoding
→ Memory decoding using PALs

Linear decoding →

Linear decoding refers to a decoding technique where address line are directly mapped to memory rows or columns using a straight forward and predictable logic pattern which is commonly used in SRAM and DRAM.

Advantage- Does not required decoding hardware

Disadvantage- ~~Bus conflict- two or more lines A10-~~

Bus conflict-

If more than one of address line (A10- A13) are low (0) at the same time, more that one SRAM get selected at once. This cause bus conflict, leading to errors.

Solution-

The software should be carefully written so that it never read or writes f address where multiple bits are low at the same time.

Memory foldback-

This happens when the same SRAM get selected for different address whi wastes large amount of address space.

Solution-

Use a fully decoded memory addressing which ensure that each address select only one specific memory address.

Didn't understand full/partial decoding

Programmable Logic device → (PLDs)

PLDs are integrated circuits that contain an array of AND gates and another array of OR gates. They are three type-

1. Programmable read only memory.
2. Programmable array logic.
3. Programmable logic array.

PROM →

PROM is a programable logic device that has fixed and array and Programmable OR array.

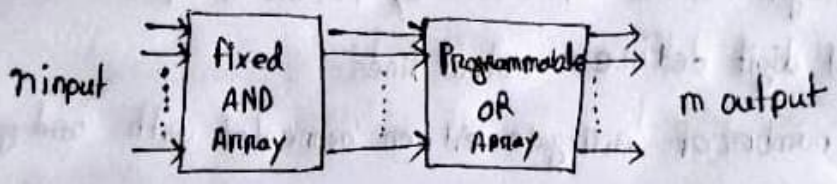


fig- PROM

PAL →

PLA is a programable logic device that has a programmable AND array and a fixed OR array.

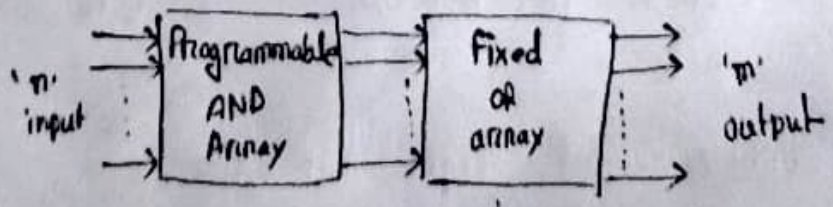


fig- PAL

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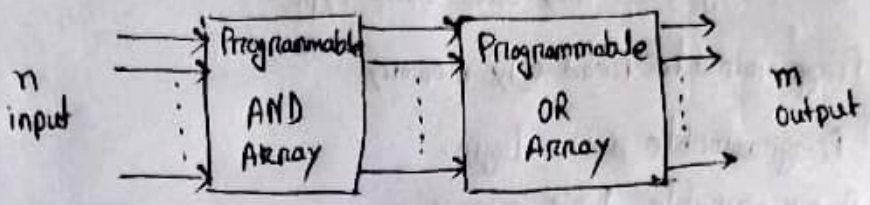


fig- PLA

PLA are indentified by two de digit number followed by a letter and then a digit.

- two digit number specified by number of input
- last digit defines output lines
- fix number of AND gate either connected with ~~and gate or nor~~ OR gate or NOR gate.

H → OR gate
L → NOR gate

Example- 10H8, 10L8

Example-1

A processor having 16-bit address line and byte addressable. Interface 4096 X8 RAM with the processor.

1. Memory capacity of the processor – $\boxed{\text{Address space} \times \text{word size}}$

$\boxed{\text{Address space} = 2^{\text{address line}}}$ $\Rightarrow 2^{16} \times$ word size

$\Rightarrow$ 65536 bytes (As byte addressable)

$\Rightarrow$ 64 kB

2. How many address required for RAM? (Address line)

Address space $= 2^{\text{address line}}$

$\Rightarrow 4096 = 2^{AL}$

$\Rightarrow 2^{12} = 2^{AL}$

$\Rightarrow AL = 12$

3. How many RAM chips are required?

$$\boxed{\text{Number of RAM} = \frac{\text{Processor capacity}}{\text{RAM capacity}}}$$

$$= \frac{2^{16}}{2^{12}}$$

$$= 2^{4} = 16$$

4. Size of Address decoder?

n input for 2^n output $\quad n = \log_2(\text{Number of chips})$

$= \log_2 16 = 4$

$\therefore$ 4 to 16 decoder needed.

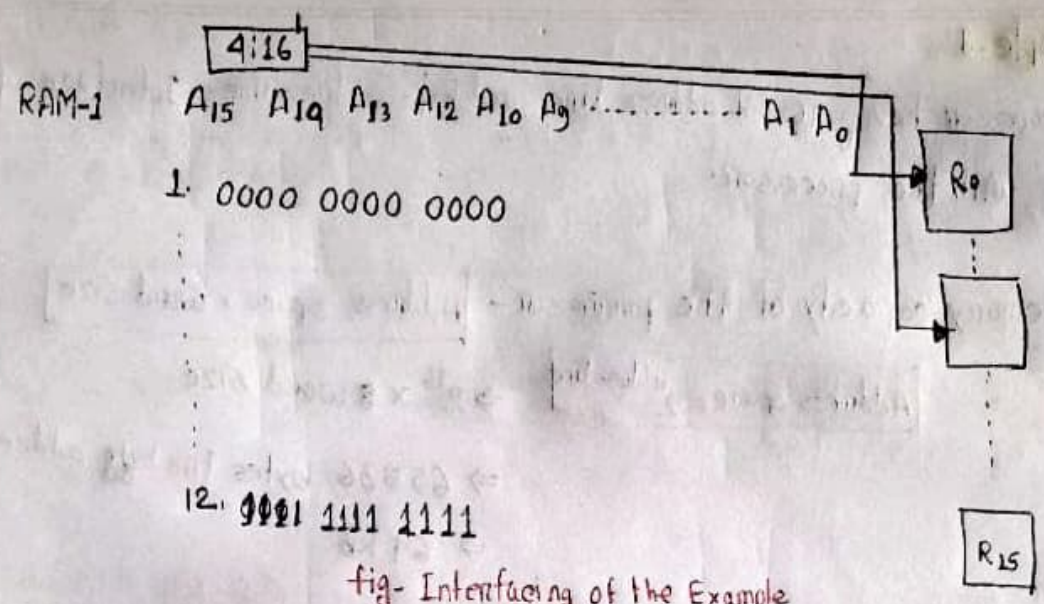


fig- Interfacing of the Example

Memory Range for RAM-1 → 0000-0FFF

first 4 bit for RAM and other for memory address

first 4 bit → Absulute address
flowed 12 bit → Partial address

Example-2

Interface two chips of 4K RAM and 1 chip of 2K ROM. starting address 8000 H.

Sol^n →

RAM-1 = 4k = 4096 bytes

RAM-2 = 4k = 4096 bytes

ROM = 2k = 2048 bytes

যেহেতাব না থাকলে byte addressable ধরে নিতে হবে।

Total memory needed = 4+4+2 = 10k bytes

given starting address 8000 H.

for RAM-1 →

starting Address - 8000H

Size = 4k = 1000 H

Ending address = 8000 H + 1000 H − 1 = 8FFF H

∴ 8000 H to 8FFF H

~~RAM 2 R~~

For RAM-2 →

starting address = 9000 H

size = 4K = 1000 H

Ending address = 9000 H + 1000 H − 1 = 9FFF H

∴ 9000 H to 9FFF H

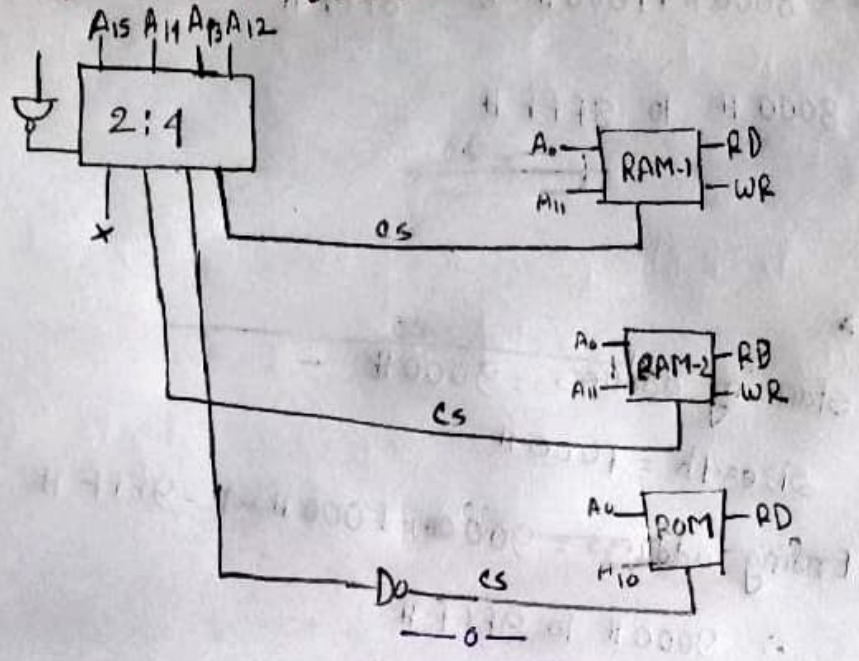
for ROM →

Starting Address = A000 H

size = 2k = 800 H

Ending Address = A000 H + 800 H − 1 = A7FF H

∴ A000 H to A7FF H

	A_{15} A_{14} A_{13} A_{12}	A_{11} A_{10} ..... A_1 A_0	IN HEX
RAM-1	1000	000 00	8000 H
	⋮	⋮	
	1000	11 11	8FFF
RAM-2	10001	00 00	9000 H
	⋮	⋮	
	1001	11 11	9FFF H
ROM	1010	0/1 0 00	A000 H → A07FF H ↳ 0 ধরলে
	⋮	⋮	
	1010	0/1 0 11	A800 H → AFFF → 1 ধরলে

Three Chips so 2 to 4 decoder.